PREVENTING TRAIN COLLISIONS



INDIAN RAILWAYS' ANTI-COLLISION SYSTEM, KAVACH, HOLDS GREAT PROMISE FOR REDUCING TRAIN ACCIDENTS. BUT ITS SLOW ROLLOUT REMAINS AN ISSUE

By Avishek G. Dastidar

IT **was sometime in March last year,** when two trains, travelling at 100 kilometres per hour, were racing towards each other from opposite sides, somewhere between Hyderabad and Lingampalli in Telangana. Just when you thought a head-on collision was inevitable, the two locomotives screeched to a halt, barely a hundred metres away from each other, as if guided by an invisible force. "Kavach," the Union railway minister Ashwini Vaishnaw, who was travelling in one of the trains, had declared, "is successful." He was referring to India's very own automatic train protection (ATP) system.

Before you could heave a sigh of relief and consign rail collisions to history, India saw at least two major train accidents this year, which together saw over 300 people being killed. On the evening of October 29, loco pilot S.M.S. Rao and his assistant S. Chiranjeevi on the Visakhapatnam-Rayagada train were caught in almost the same situation as Vaishnaw witnessed, except that this was no drill, and there was no Kavach. Jumping two successive signals, Rao and Chiranjeevi, quite inexplicably, drove their train, travelling at 60 kmph, straight into another stationary passenger train. Both died on impact, as did 12 passengers. A preliminary probe blamed the deceased engine drivers and said the signals on the route had been faulty, leading to a mix-up.

Just months before that, on June 2, another train tragedy unfolded in the Balasore district of Odisha, when the 12841 Coromandel Express entered a loop line it was not meant to be on, ramming into a goods train. As many as 21 coaches of the passenger train derailed, three of which collided with another oncoming train on the adjacent track. Around 295 people were killed and over 1,200 injured. Later, a probe found that faulty wiring at the signalling station had led to the disaster.

It was to rid the railways of mishaps such as these—as well as several other near-misses—that India conceptualised its ATP system as early as 2012. Over a decade later, only around 3,000 km of the busiest Delhi-Mumbai and Delhi-Howrah routes will have been fitted with Kavach by 2024. As of now, it covers just 1,465 km—or 2.1 per cent—of India's 68,000 km rail network.

Vaishnaw willingly concedes India's laggardness in this respect. "The entire world moved to ATP systems as early as the 1990s. In fact, bigger railways moved to this system in the late '80s. India had to wait till 2016, for the arrival of PM Narendra Modi to start focusing on the core safety of railways," he told INDIA TODAY. The minister has now directed the railways to expedite the rollout. "We have taken it up on a mission mode. Kavach will be installed in the same way that railway electrification is happening," he says.

THE TECH SUCCESS

Originally known as the Train Collision Avoidance System, Kavach, which translates as "armour", has taken India over a decade to perfect. Hopes for it are running high. Kavach brings line-side signals—green, yellow and red—into the loco pilot's engine cabin, doing away with the need for the driver to actually look out for signals on the tracks. More crucially, however, the system can sense if the driver is about to jump a signal and applies brakes automatically via the anti-collision system installed